CADENA DE TÉRMINOS DOMINANTES

Sean $k \in \mathbb{R}, a, p \in \mathbb{R}^+, a > 1$ entonces para n suficientemente grande se tiene que:

$$k \ll \ln n \ll n^p \ll a^n \ll n! \ll n^n$$

$$\lim_{x \to +\infty} \frac{f(x)}{g(x)} \quad , \quad f(x) > g(x) = +\infty \quad , \quad f(x) < g(x) = 0$$

$$\lim_{x \to +\infty} \frac{\ln(x)}{n^n} = 0$$

$x+3 = x$

K sumadas $= 0$

K multiplicadas $= 1$

$$\lim_{x \to +\infty} \frac{n^n}{\ln(x)} = +\infty$$

$x+3 = x \qquad 3x+3 = x$

Criterio de la razon

La serie $\sum_{n=p}^{\infty}$ Que tenga ! o ...

Converge si

$$\lim_{n \to +\infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$$

Diverge si

$$\lim_{n \to +\infty} \left| \frac{a_{n+1}}{a_n} \right| > 1 \quad 0$$

∞

$$\sum_{n=0}^{\infty} \frac{1\cdot 4\cdot 7\cdot (3n+1)}{2^{n+1}\,(n+1)!}$$

Paso 1: Definir a_{n+1}

$$a_{n+1} = \frac{1\cdot 4\cdot 7\cdot (3n+1)\cdot(3(n+1)+1)}{2^{(n+1)+1}\cdot((n+1)+1)!}$$

Paso 2: Definir a_n (original)

$$a_n = \frac{1\cdot 4\cdot 7\cdot (3n+1)}{2^{n+1}\,(n+1)!}$$

Paso 3: Poner $\lim\limits_{n\to+\infty} \frac{a_{n+1}}{a_n}$

$$\lim_{n\to+\infty} \frac{\dfrac{1\cdot 4\cdot 7\cdot (3n+1)(3(n+1)+1)}{2^{(n+1)+1}\cdot((n+1)+1)!}}{\dfrac{1\cdot 4\cdot 7\cdot (3n+1)}{2^{n+1}\,(n+1)!}}$$

$3(n+1)+1$

$3n+3+1 = 3n+4$

$$\lim_{n\to+\infty} \frac{\dfrac{\cancel{1\cdot 4\cdot 7\cdots(3n+1)}\,(3n+4)}{2^{n+2}\,(n+2)!}}{\dfrac{\cancel{1\cdot 4\cdot 7\cdot(3n+1)}}{2^{n+1}\,(n+1)!}}$$

$$\frac{\dfrac{\cancel{a}}{b}}{\dfrac{\cancel{a}}{c}} \qquad \frac{\dfrac{b}{\cancel{a}}}{\dfrac{c}{\cancel{a}}}$$

$$\lim_{n\to+\infty} \frac{(3n+4)\,2^{n+1}\,(n+1)!}{2^{n+2}\,(n+2)!}$$

$$\lim_{n\to+\infty} \frac{(3n+4)\,2^{n+1}\,(n+1)!}{2^{n+2}\,(n+2)!}$$

$$\lim_{n\to+\infty} \frac{(3n+4)\cdot\cancel{2^{n+1}}\cdot\cancel{(n+1)!}}{\cancel{2^{n+1}}\cdot 2\,(n+2)\,\cancel{(n+1)!}}$$

$$\lim_{n\to+\infty} \frac{3n+\cancel{4}}{2n+\cancel{4}}$$

$$\lim_{n\to+\infty} \frac{3\cancel{n}}{2\cancel{n}} = \frac{3}{2} = 1.5 > 1 \quad \therefore \boxed{\text{Diverge}}$$

!!!!

$$\sum_{n=3}^{\infty} \frac{(3n+5)!}{20n-7} \qquad \lim_{n\to+\infty} \frac{a_{n+1}}{a_n}$$

$$\lim_{n\to+\infty} \frac{\dfrac{(3n+8)!}{20n+13}}{\dfrac{(3n+5)!}{20n-7}} \qquad a_{n+1} = \frac{(3(n+1)+5)!}{20(n+1)-7}$$

$$\frac{(3n+3+5)!}{20n+20-7}$$

$$\lim_{n\to+\infty} \frac{(3n+8)!\,(20n-7)}{(3n+5)!\,(20n+13)} \qquad \frac{(3n+8)!}{20n+13}$$

$$\lim_{n\to+\infty} \frac{(3n+8)(3n+7)(3n+6)\cancel{(3n+5)!}(20n-7)}{\cancel{(3n+5)!}(20n+13)}$$

$$\lim_{n\to+\infty} \frac{(3n+8)(3n+7)(3n+6)(20n-7)}{(20n+13)}$$

$$K \cdot +\infty = +\infty$$
$$K + \infty = +\infty$$

$$\lim_{n\to+\infty} \frac{n \cdot n \cdot n \cdot \cancel{n}}{\cancel{n}}$$, cadena de dominancia

$$n^3 = +\infty^3 = +\infty$$

$\therefore$ Diverge

$$n! = n(n-1)(n-2)(n-3)\dots!$$

$$(3n+8)! = (3n+8)(3n+8-1)(3n+8-2)(3n+8-3)!$$
$$(3n+8)(3n+7)(3n+6)(3n+5)!$$

$$\sum_{n=1}^{\infty} \frac{(n+1)\cdot 2^n}{n!}$$

$\lim_{n\to+\infty} \frac{a_{n+1}}{a_n}$ > 1 ✓ Conver

< 1 ✗ Diver

0 $+\infty$

$$a_{n+1} = \frac{[(n+1)+1]\cdot 2^{n+1}}{(n+1)!} \qquad a_n = \frac{(n+1)\cdot 2^n}{n!}$$

$$a_{n+1} = \frac{(n+2)\cdot 2^{n+1}}{(n+1)!}$$

$$\lim_{n\to+\infty} \frac{\dfrac{(n+2)\cdot 2^{n+1}}{(n+1)!}}{\dfrac{(n+1)\cdot 2^n}{n!}}$$

$$\lim_{n\to+\infty} \frac{(n+2)\cdot 2^{n+1}\quad n!}{\boxed{(n+1)!}\,(n+1)\cdot 2^n} \qquad \frac{2^{n+1}}{2^n} = \frac{\cancel{2^n}\cdot 2}{\cancel{2^n}} = 2$$

$$(n+1)! = (n+1)\cdot n!$$

$$\lim_{n\to+\infty} \frac{(n+2)\cdot \cancel{2^n}\cdot 2\cdot \cancel{n!}}{(n+1)^1\cdot \cancel{n!}\cdot (n+1)^1\cdot \cancel{2^n}}$$

$$\lim_{n\to+\infty} \frac{\cancel{2}^{\,1}(n+\cancel{2}^{\,0})}{(n+\cancel{1}^{\,0})^2} \qquad \cancel{\pi}^{\,1}\cdot x = x \qquad x+\cancel{\pi}^{\,0} = x$$

$$\lim_{n\to+\infty} \frac{\cancel{n}}{\cancel{n^2}} \to \lim_{n\to+\infty} \frac{1}{n} = 0 \qquad \frac{\pi}{+\infty} = 0$$

$\to 0 < 1$ ∴ $\boxed{\text{Diverge}}$

$$\frac{n}{(n^n + n!)(\ln(n) + 2)}$$

$\hookrightarrow n! < n^n \qquad \hookrightarrow \pi < \ln(n)$

$$\frac{n}{n^n \cdot \ln(n)}$$

$$\sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdot 6 \cdots (2n)}{n!}$$

$a_n = 2 \cdot 4 \cdot 6 \cdots (2n)$

$a_{n+1} = 2 \cdot 4 \cdot 6 \cdots (2n)(2n(n+1))$

$= 2 \cdot 4 \cdot 6 \cdots (2n)(2n+2)$

$$\lim_{n\to+\infty} \frac{\frac{\cancel{2 \cdot 4 \cdot 6 \cdots (2n)}(2n+2)}{(n+1)!}}{\frac{\cancel{2 \cdot 4 \cdot 6 \cdots (2n)}}{n!}}$$

$$\frac{\frac{\cancel{a}}{b}}{\frac{\cancel{a}}{c}} \qquad \frac{b}{\cancel{a}} \cdot \frac{c}{\cancel{a}}$$

$$\lim_{n\to+\infty} \frac{(2n+2) \cdot \cancel{n!}}{(n+1) \cdot \cancel{n!}} \rightarrow (n+1)! = (n+1) \cdot n!$$

$$\lim_{n\to+\infty} \frac{2n+2}{n+1}$$

$$\lim_{n\to+\infty} \frac{2\cancel{n}}{\cancel{n}} = \boxed{2 > 1 \quad \text{is Diverge}}$$

$$\sum_{n=0}^{\infty} \frac{[1\cdot 3\cdot 5 \cdots (2n-1)]\cdot n!}{(2n+1)!}$$

$2(n+1)-1 = 2n+2-1$
$= 2n+1$

$$a_{n+1} = \frac{[1\cdot 3\cdot 5 \cdots (2n-1)(2n+1)]\cdot (n+1)!}{(2n+3)!}$$

$(2(n+1)+1)! = (2n+2+1)!$
$= (2n+3)!$

$$a_n = \frac{[1\cdot 3\cdot 5 \cdots (2n-1)]\cdot n!}{(2n+1)!}$$

$$\lim_{n\to+\infty} \frac{\dfrac{[\cancel{1\cdot 3\cdot 5 \cdots (2n-1)}(2n+1)]\cdot (n+1)!}{(2n+3)!}}{\dfrac{[\cancel{1\cdot 3\cdot 5 \cdots (2n-1)}]\cdot n!}{(2n+1)!}}$$

$$\lim_{n\to+\infty} \frac{(2n+1)(n+1)n!\,(2n+1)!}{(2n+3)!\,n!}$$

$$\lim_{n\to+\infty} \frac{(2n+1)\cancel{(2n+1)!}(n+1)\cdot\cancel{n!}}{(2n+3)\underbrace{(2n+2)}\cancel{(2n+1)!}\cdot\cancel{n!}}$$

$$\lim_{n\to+\infty} \frac{(2n+1)\cancel{(n+1)}}{(2n+3)\,2\cancel{(n+1)}}$$

$$\lim_{n\to+\infty} \frac{\cancel{2n}}{\cancel{4n}} = \boxed{\frac{1}{2} < 1 \quad \therefore \text{Converge}}$$

Criterio de la raiz

Todas las que se pueden resolver por criterio de la raiz se pueden resolver por el criterio de la razon

Criterio de la raiz

La serie $\sum_{n=p}^{\infty} (a_n)^n$ ← forma

No p

Converge si $\lim_{n\to+\infty} \sqrt[n]{|a_n|} < 1$

Diverge si $\lim_{n\to+\infty} \sqrt[n]{|a_n|} > 1$ o ∞

$$\lim_{n\to+\infty} \sqrt[n]{k} = 1, \quad k \text{ constante}$$

$$\lim_{n\to+\infty} \sqrt[n]{\frac{k}{n}} = 1, \quad k \text{ una constante}$$

$$\lim_{n\to+\infty} \sqrt[n]{P(x)} = 1, \quad P(x) \text{ un polinomio}$$

$$\lim_{n\to+\infty} \sqrt[n]{\frac{P(x)}{Q(x)}} = 1, \quad P(x), Q(x) \text{ polinomios}$$

$$\sqrt[n]{n^n} = n$$

$$\sum_{n=1}^{\infty} \left(\frac{3n^2+3n+1}{5n^2+5} \right)^{3n}$$

$$(x^2)^n = x^{2n}$$

$$\lim_{n \to +\infty} \sqrt[n]{\left(\frac{3n^2+3n+1}{5n^2+5} \right)^{3n}}$$

$$x^{2n} = (x^n)^2 = (x^2)^n$$

$$\lim_{n \to +\infty} \sqrt[\cancel{n}]{\left[\left(\frac{3n^2+3n+1}{5n^2+5} \right)^3 \right]^{\cancel{n}}}$$

$$\sqrt[n]{x \cdot y} = \sqrt[n]{x} \cdot \sqrt[n]{y}$$

$$\sqrt[n]{n^n} = n$$

$$\lim_{n \to +\infty} \left(\frac{3n^2+3n+1}{5n^2+5} \right)^3$$

$$\lim_{n \to +\infty} \left(\frac{3\cancel{n^2}}{5\cancel{n^2}} \right)^3 = \left(\frac{3}{5} \right)^3 = \frac{3^3}{5^3} = \frac{9}{25} < 1$$

$$\boxed{\therefore \text{ Divergente}}$$